

Generative Calculus: Resource-Scaled Operational Envelopes, Reachable Capability, and the Limits of Apparent Expansion

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Abstract

Scientific theories typically compare states, experiments, resources, or algorithms inside a prespecified operational setting. We develop Generative Calculus, a framework for comparing complete resource-scaled envelopes of physically realizable capability. An embodied operational world specifies task families, protocols, observable error, composition, and vector-valued resource costs; its total envelope records joint task-scale-error-budget feasibility. We prove an envelope representation theorem, generative preorder and equivalence results, exact Blackwell-Le Cam and resource-theory reductions, computational formal-versus-feasible separation and hierarchy results, Turing-presented undecidability, and marginal incompleteness. We strengthen the framework with a proper-projection irreducibility theorem whose parity construction agrees on every strict coordinate projection while differing globally, yielding an unbounded generative-order hierarchy. For compact convex envelope trajectories, support functions give a representation-invariant reconstruction theorem from its support-function derivative. A bijective cyclic construction shows that microscopic bijectivity need not imply scalable operational reversibility. We replace unrestricted envelope conservation by a proved budgeted reachable-envelope transport law: capabilities reachable after an internal transition were already reachable before once transition cost is charged. Reproducible software validates the theory against Blackwell comparison, communication complexity, reachability, synthetic stress tests, and public MLPerf Power data. The framework claims scoped mathematical synthesis only, not established new fundamental physics, and identifies precise boundaries for future falsification.

Keywords: operational envelopes; resource-scaled simulation; reachability; resource theories; statistical experiments; physical computation

1 Introduction

A mature physical or computational theory normally fixes a language of states, admissible transformations, measurements, and costs, then asks what follows inside that setting. This pattern is extraordinarily successful, but it obscures a different comparison problem: two worlds may implement the same named task while occupying different *joint capability regions* once accuracy, scale, time, energy, memory, communication, precision, coherence, and other resources are treated simultaneously. Generative Calculus is proposed as a mathematical language for these resource-scaled capability regions rather than as a replacement for mechanics, information theory, or resource theory.

The intellectual lineage is intentionally broad. Statistical comparison of experiments establishes exact and approximate simulation orders (Blackwell, 1951, 1953; Khan & Yu, 2025; Le Cam, 1964; Le Cam & Yang, 2000; Strasser, 1985; Torgersen, 1991; Wald, 1950); information theory supplies operational measures and asymptotic reasoning (Cover & Thomas, 2006; Csiszár, 1967; Kullback & Leibler, 1951; Pinsker, 1964; Polyanskiy & Wu, 2025; Shannon, 1948; Tsybakov,

2009; van der Vaart, 1998); and majorization and convex analysis provide canonical partial-order tools (Boyd & Vandenberghe, 2004; G. H. Hardy et al., 1934; Hiriart-Urruty & Lemaréchal, 2001; Marshall et al., 2011; Nielsen, 1999; Rockafellar, 1970; Rockafellar & Wets, 1998).

Quantum resource theory clarifies how allowed operations, monotones, and asymptotic rates organize physical convertibility (Brandão & Gour, 2015; Brandão et al., 2015; Chitambar & Gour, 2019; Coecke et al., 2016; Fritz, 2017; Gour et al., 2015; Hayashi & Yamasaki, 2025; M. Horodecki & Oppenheim, 2013; R. Horodecki et al., 2009; Khanian et al., 2023; Liu et al., 2024; Lostaglio, 2019; Luo, 2025; Regula, 2018; Regula & Lami, 2024; Takagi & Regula, 2019). General probabilistic and operational approaches show that states and measurements can be compared without presupposing quantum mechanics (Barnum et al., 2007, 2010, 2014; Barrett, 2007; Buscemi, 2012; Catani et al., 2026; Chiribella et al., 2010, 2011; L. Hardy, 2001; Heinosaari et al., 2016; Ioannou et al., 2022; Janotta & Hinrichsen, 2014; Lee & Barrett, 2015; Plávala, 2021; Schmid et al., 2018; Short & Wehner, 2010; Spekkens, 2005).

The physical-computation lineage makes a second boundary explicit: formal describability is not identical to feasible physical realization (Aaronson, 2005; Bennett, 1973, 1982; Deutsch, 1985, 2013; Deutsch & Marletto, 2015; Feynman, 1982; Goold et al., 2016; Landauer, 1961; Lloyd, 2000; Margolus & Levitin, 1998; Marletto, 2015, 2021; Parrondo et al., 2015; Reeb & Wolf, 2014; Wolpert, 2019). Complexity theory then supplies unconditional separations and undecidability tools (Arora & Barak, 2009; Babai & Kimmel, 1997; Blum, 1967; Buhrman et al., 2010; Chaitin, 1974; Church, 1936; Cook, 1971; Goldreich, 2008; Hartmanis & Stearns, 1965; Hopcroft et al., 2006; Karp, 1972; Kushilevitz & Nisan, 1997; Ladner, 1975; Li & Vitányi, 2008; Newman, 1991; Odifreddi, 1989; Papadimitriou, 1994; Rice, 1953; Rogers, 1987; Sipser, 2012; Turing, 1936; Yao, 1979).

Finally, Pareto analysis, category theory, reflection, and open-ended evolution demonstrate why several tempting novelty claims must be rejected. Multiobjective trade-offs are classical (Deb, 2001; Ehrgott, 2005; Miettinen, 1999; Pareto, 1896; Zitzler & Thiele, 1999); compositional structures are deeply developed (Abramsky & Coecke, 2004; Awodey, 2010; Baez & Pollard, 2017; Baez & Stay, 2011; Coecke & Duncan, 2011; Coecke & Kissinger, 2017; Fong, 2015; Fong & Spivak, 2019; Lawvere & Schanuel, 2009; Mac Lane, 1998; Selinger, 2011); reflective rule change is established in rewriting logic (Clavel et al., 2007; Meseguer, 1992); and continual, expansive, or transformational novelty is already central to artificial-life research (Adams et al., 2017; Bedau et al., 2000; Borg et al., 2024; Channon, 2024; Holland, 1992; S. Kauffman & Roli, 2021; S. A. Kauffman, 2000; Langton, 1986; Lenski et al., 2003; Maynard Smith, 1982; Mesoudi & Thornton, 2018; Packard et al., 2019a, 2019b; Sayama & Nehaniv, 2025; Taylor, 2019; Taylor et al., 2016; Yang, 2025).

The contribution is consequently narrow. We do *not* claim novelty for changing state spaces, meta-rules, resource vectors, Pareto fronts, experiment simulability, or possible/impossible transformations. We instead formalize a *whole embodied operational envelope*: the set of task-scale-error-budget tuples jointly achievable by a world, together with translations that must preserve the operational structure of entire task families. The framework provides a common container in which classical comparison results appear as restricted projections, while an internal result—marginal incompleteness—shows that coordinate-wise resource optima need not reconstruct the joint envelope.

2 Positioning and novelty boundary

Table 1 summarizes the novelty audit before any theorem is stated. This table is deliberately conservative: an “original target” means that the present formulation is the paper’s research contribution, not that universal historical priority has been proved.

Table 1: Novelty boundary after collision testing against adjacent literatures.

Component	Status here	Nearest established territory
Changing state spaces, laws, or meta-rules	Not claimed	Evolving spaces, reflective rewriting, open-ended evolution
Experiment simulability and deficiency	Reduction only	Blackwell and Le Cam comparison
Free/resource convertibility	Reduction only	Resource theories and asymptotic rates
Multi-resource Pareto structure	Building block	Multiobjective optimization and multi-resource thermodynamics
Whole task-scale-error- budget-composition envelope	Core scoped contribution	Operational theories, resource theories, reachability
Proper-projection irre- ducibility	Proved GC theorem	Marginal consistency and higher-order interaction phenomena
Convex support-function re- construction	Proved restricted theorem	Convex geometry and vector-valued calculus
Budgeted reachable- envelope transport	Proved GC structural theorem	Reachability/viability transitivity
Unrestricted envelope con- servation	Rejected	Counterexample given in this paper
Experimentally new law of nature	Not claimed	Requires independent prediction and confirmation

3 Embodied operational worlds

Definition 1 (Embodied operational world). An embodied operational world is a tuple

$$W = (\mathcal{T}, \mathcal{P}, \mathcal{R}, c, \text{Pr}, \circ),$$

where $\mathcal{T}$ is a collection of task families, $\mathcal{P}$ is the set of admissible protocols, $\mathcal{R} = \{R_1, \dots, R_d\}$ is a finite resource coordinate set, $c(P; n, \epsilon) \in \mathbb{R}_+^d$ is the protocol cost vector, Pr specifies observable success statistics, and $\circ$ specifies admissible composition.

For a task T , scale n , and error tolerance ϵ , let

$$C_W(T; n, \epsilon) = \{c(P; n, \epsilon) : P \text{ realizes } T\}.$$

We use the coordinatewise order $x \preceq y$ when $x_i \leq y_i$ for every resource coordinate.

Definition 2 (Embodied operational envelope). For budget $B \in \mathbb{R}_+^d$,

$$\mathcal{E}_W(n, \epsilon, B) = \{T : \exists c \in C_W(T; n, \epsilon), c \preceq B\}.$$

The total envelope is

$$\mathcal{E}(W) = \{(T, n, \epsilon, B) : T \in \mathcal{E}_W(n, \epsilon, B)\}.$$

Definition 3 (Capability frontier). The Pareto frontier of task T is

$$\Phi_W(T; n, \epsilon) = \text{Min}_{\preceq} C_W(T; n, \epsilon).$$

Algorithm 1 constructs the finite Pareto frontier using the same logic implemented in the software package.

Algorithm 1 Finite embodied-envelope frontier construction

Require: Finite achievable cost set $C \subset \mathbb{R}_+^d$

Ensure: Pareto frontier Φ

```
1:  $\Phi \leftarrow \emptyset$ 
2: for all  $p \in C$  do
3:   if there is no  $q \in C$  with  $q \neq p$  and  $q \preceq p$  then
4:      $\Phi \leftarrow \Phi \cup \{p\}$ 
5:   end if
6: end for
7: return  $\Phi$ 
```

4 Representation and translation structure

Theorem 4 (Envelope Representation Theorem). *Assume $C_W(T; n, \epsilon)$ is compact whenever T is realizable. Then*

$$T \in \mathcal{E}_W(n, \epsilon, B) \iff \exists \phi \in \Phi_W(T; n, \epsilon) \text{ with } \phi \preceq B.$$

Proof. The reverse implication is immediate because every frontier point is achievable. For the forward implication choose any achievable $c \preceq B$. The compact set $K = C_W(T; n, \epsilon) \cap \{x : x \preceq c\}$ is nonempty. A minimizer of $f(x) = \sum_i x_i$ on K exists. If another achievable point strictly dominated that minimizer, its f value would be smaller, a contradiction. Hence a Pareto-minimal $\phi \preceq c \preceq B$ exists. $\square$

Definition 5 (Resource-scaled operational translation). For an admissible overhead class $\mathcal{H}$, write

$$B \preceq_G^{\mathcal{H}} A$$

when every task family represented in B admits an operationally faithful realization in A whose error transformation and resource distortion lie in $\mathcal{H}$ and whose compositions preserve the designated operational relations.

Theorem 6 (Generative Preorder and Equivalence Theorem). *If $\mathcal{H}$ contains identity overhead and is closed under composition, then $\preceq_G^{\mathcal{H}}$ is a preorder. Mutual reachability,*

$$A \simeq_G^{\mathcal{H}} B \iff A \preceq_G^{\mathcal{H}} B \wedge B \preceq_G^{\mathcal{H}} A,$$

is an equivalence relation.

Proof. Identity translation establishes reflexivity. Composition of faithful translations establishes transitivity because $\mathcal{H}$ is composition-closed. The induced mutual-reachability relation is reflexive, symmetric by definition, and transitive by the preorder's transitivity. $\square$

Algorithm 2 gives the finite translation audit used in the experiments; it makes no claim of deciding the general Turing-presented problem.

5 Reductions to established theories

The reduction results are intentionally framed as consistency checks rather than novelty claims. Under statistical-experiment specialization, the GC translation order should coincide exactly with established comparison theory.

Algorithm 2 Finite whole-envelope inclusion audit

Require: Worlds A, B , finite budget grid $\mathcal{B}$, scale/error settings

Ensure: Boolean inclusion verdict and witnesses

```
1:  $V \leftarrow \emptyset$ 
2: for all  $b \in \mathcal{B}$  do
3:   Compute  $\mathcal{E}_A(b)$  and  $\mathcal{E}_B(b)$  from Pareto frontiers
4:   for all  $T \in \mathcal{E}_B(b) \setminus \mathcal{E}_A(b)$  do
5:      $V \leftarrow V \cup \{(T, b)\}$ 
6:   end for
7: end for
8: return  $(V = \emptyset, V)$ 
```

Theorem 7 (Blackwell–Le Cam Reduction). *Restrict worlds to statistical experiments with a common parameter space and restrict translations to Markov kernels. Exact $\preceq_G$ is Blackwell garbling. If the translation loss is*

$$\delta_G(F \mid E) = \inf_K \sup_{\theta} d_{\text{TV}}(K P_{\theta}, Q_{\theta}),$$

then δ_G is the corresponding Le Cam deficiency under the same conventions.

Proof. Both statements follow by identity of admissible translation and loss definitions with the specialized comparison-of-experiments problem (Blackwell, 1953; Le Cam & Yang, 2000; Torgersen, 1991). $\square$

Theorem 8 (Resource-Theory Reduction). *Let a resource theory consist of objects and free transformations $F(X, Y)$. Construct tasks $T_{X \rightarrow Y}$ and set zero-cost realizability exactly when $F(X, Y)$ is nonempty. Then GC reachability restricted to these tasks coincides with resource-theoretic convertibility; the many-copy specialization yields the same asymptotic conversion-rate optimization.*

Proof. The map is definitional: a task is feasible exactly when the corresponding free transformation exists. Tensor powers and asymptotic rate optimization preserve the same feasible conversions (Brandão & Gour, 2015; Chitambar & Gour, 2019; Coecke et al., 2016). $\square$

6 Hierarchy, computability, and dynamic structure

The next results distinguish what is inherited from established mathematics from what is specific to the envelope formulation. Computational hierarchy and undecidability are used as exact specializations rather than presented as new complexity results.

Theorem 9 (Formal/Feasible Separation, computational specialization). *There exist task families formally describable in a universal computational language that are not realizable inside a stricter time envelope. In particular, for time-constructible bounds satisfying the deterministic time-hierarchy separation, a language in $\text{DTIME}(n^{k+1}) \setminus \text{DTIME}(n^k)$ is formally Turing-describable but absent from the $O(n^k)$ embodied time envelope.*

Proof. The description is a finite Turing-machine program. The deterministic time-hierarchy theorem provides a language requiring asymptotically more than the lower time bound (Arora & Barak, 2009; Hartmanis & Stearns, 1965). Thus formal representation does not imply membership in the specified resource-bounded envelope. $\square$

Corollary 10 (Infinite computational envelope hierarchy). *For $W_k = \text{DTIME}(n^k)$, overhead criteria that preserve the hierarchy induce an infinite strict sequence of time-bounded capability envelopes.*

Algorithm 3 Marginal-incompleteness stress test

Require: Low cost ℓ , high cost $h \geq \ell$, joint budget $b = (\ell, \ell)$

- 1: $\Phi_A \leftarrow \{(\ell, h), (h, \ell)\}; \Phi_B \leftarrow \{(\ell, \ell)\}$
 - 2: Compute coordinate minima m_A, m_B
 - 3: Evaluate feasibility of b against both frontiers
 - 4: Compute $g = \min_{p \in \Phi_A} \|p - m_A\|_2$
 - 5: **return** $(m_A = m_B, b \notin \mathcal{E}_A, b \in \mathcal{E}_B, g)$
-

Theorem 11 (Metric Generative Derivative). *Let the quotient of generative-equivalence classes be equipped with a metric d_G , and let $\gamma : [0, T] \rightarrow (\mathfrak{E}_G, d_G)$ be absolutely continuous. Then the metric derivative*

$$|D_G W|(t) = \lim_{h \rightarrow 0} \frac{d_G(\gamma(t+h), \gamma(t))}{|h|}$$

exists for almost every t .

Proof. This is the standard metric-derivative result for absolutely continuous curves in metric spaces, applied after a legitimate metric has been specified on the GC quotient (Ambrosio et al., 2008; Burago et al., 2001; Villani, 2009). $\square$

Theorem 12 (Undecidability of Turing-presented envelope containment). *For arbitrary Turing-presented worlds whose task languages are recursively enumerable, general envelope containment and equality are undecidable.*

Proof. Fix a world whose task language is empty. A decision procedure for containment in that world would decide whether the language of an arbitrary Turing machine is empty; equality with the empty world would do the same. These are nontrivial semantic properties and contradict standard computability results (Rice, 1953; Rogers, 1987; Turing, 1936). $\square$

7 Compatibility geometry beyond coordinate summaries

The first GC-specific result concerns what is lost when a whole feasibility relation is collapsed to separately optimized coordinates. Multiobjective trade-offs themselves are classical (Deb, 2001; Ehrgott, 2005; Miettinen, 1999; Pareto, 1896); the point here is to identify the information that must be retained by an embodied operational envelope.

Theorem 13 (Marginal Incompleteness). *For $d \geq 2$, coordinate-wise minima of a Pareto frontier do not in general determine the embodied operational envelope.*

Proof. Let World A have frontier $\{(1, 3), (3, 1)\}$ and World B have frontier $\{(1, 1)\}$. Both have coordinate minima $(1, 1)$. Under joint budget $(1, 1)$, however, B realizes the task and A does not. Hence identical coordinate minima coexist with distinct envelopes. $\square$

Corollary 14 (Coordinate-summary reconstruction failure). *Any summary depending only on independently optimized coordinate minima assigns the same value to the two worlds above and therefore cannot be complete for joint-envelope reconstruction.*

Algorithm 3 operationalizes this counterexample before its numerical results are reported.

The stronger construction below tests every strict coordinate projection, not merely one-dimensional minima. Similar local-to-global and marginal-consistency phenomena occur in probability, contextuality, and relational structures; accordingly, the combinatorial skeleton is not claimed as historically new (Catani et al., 2026; Ioannou et al., 2022; Schmid et al., 2018; Spekkens, 2005).

Algorithm 4 Proper-projection irreducibility audit

Require: Dimension $m \geq 2$

- 1: Build even-parity envelope E^+ and odd-parity envelope E^-
 - 2: **for** $r = 1, \dots, m$ **do**
 - 3: **for all** $S \subseteq \{1, \dots, m\}$ with $|S| = r$ **do**
 - 4: Compare $\pi_S(E^+)$ and $\pi_S(E^-)$
 - 5: **end for**
 - 6: **end for**
 - 7: **return** the first order r with a differing projection
-

Theorem 15 (Strong Proper-Projection Irreducibility). *For every integer $m \geq 2$, there exist finite envelopes $\mathcal{E}_m^+$ and $\mathcal{E}_m^- \subseteq \{0, 1\}^m$ such that every proper coordinate projection is identical,*

$$\pi_S(\mathcal{E}_m^+) = \pi_S(\mathcal{E}_m^-), \quad S \subsetneq \{1, \dots, m\},$$

yet $\mathcal{E}_m^+ \cap \mathcal{E}_m^- = \emptyset$.

Proof. Let $\mathcal{E}_m^+$ contain the even-parity bit strings and $\mathcal{E}_m^-$ the odd-parity strings. Fix any proper coordinate set S and any assignment y on S . At least one coordinate is omitted. Choosing one omitted bit appropriately extends y to even parity; flipping that bit extends the same y to odd parity. Hence both projections equal the full cube $\{0, 1\}^{|S|}$. The two whole relations are disjoint because a bit string cannot have both parities. $\square$

Definition 16 (Pairwise GC projection order). For finite envelopes $A, B \subseteq \prod_{i=1}^m X_i$, define

$$r_G(A, B) = \min\{|S| : \pi_S(A) \neq \pi_S(B)\}.$$

Corollary 17 (Unbounded GC projection order). *For the parity pair above, $r_G(\mathcal{E}_m^+, \mathcal{E}_m^-) = m$. Consequently GC projection order is unbounded over finite envelope families.*

Proof. Every projection of order below m agrees by the theorem, while the full m -coordinate relations differ. $\square$

Algorithm 4 audits all coordinate subsets for the finite parity family before the corresponding result is shown.

8 A restricted Fundamental-Theorem sector

A fully general derivative of arbitrary operational envelopes would be unjustified. A rigorous local-to-global reconstruction is nevertheless available for compact convex trajectories when the support-function path is differentiable. Support functions uniquely determine compact convex bodies and linearize Minkowski addition (Boyd & Vandenberghe, 2004; Rockafellar, 1970; Schneider, 2014).

Theorem 18 (Convex Generative Reconstruction). *Let $K_t \subset \mathbb{R}^d$, $t \in [0, T]$, be nonempty compact convex bodies. Let $h_t(u) = \sup_{x \in K_t} \langle u, x \rangle$ for $u \in S^{d-1}$, and assume $t \mapsto h_t$ is continuously differentiable in the sup norm. Then*

$$h_t = h_0 + \int_0^t \dot{h}_s ds,$$

and K_t is uniquely determined by the right-hand side. Thus the local derivative of the support representation reconstructs the complete convex envelope trajectory.

Proof. The fundamental theorem of calculus in the Banach space $C(S^{d-1})$ gives the integral identity under the stated C^1 assumption. The support-function map is injective on nonempty compact convex bodies, so the reconstructed h_t determines exactly one K_t (Schneider, 2014). $\square$

Corollary 19 (Minkowski-path reconstruction). *For $K_t = (1 - t)K_0 + tK_1$ on $t \in [0, 1]$,*

$$h_t = (1 - t)h_0 + th_1, \quad \dot{h}_t = h_1 - h_0,$$

so the reconstruction is exact.

This theorem justifies the word *calculus* on a nontrivial restricted sector while making no claim that support-function calculus itself is new.

9 Reachable capability, reversibility, and apparent expansion

Reachability and viability theory already study sets obtainable under dynamical constraints (Aubin, 2005; Siefert et al., 2025). GC lifts the same compositional logic from state reachability to task feasibility under explicit budgets.

Definition 20 (Reachable task envelope). For a state x of a fixed operational world and budget $B \in \mathbb{R}_+^d$, let $\mathcal{R}_x(B)$ be the set of tasks realizable from x by an admissible protocol whose total resource cost is coordinate-wise at most B .

Theorem 21 (Budgeted Reachable-Envelope Transport). *Assume nonnegative resource costs and subadditive composition,*

$$c(P \circ Q) \preceq c(P) + c(Q).$$

If y is reachable from x with cost vector c_{xy} , then for every budget B ,

$$\boxed{\mathcal{R}_y(B) \subseteq \mathcal{R}_x(B + c_{xy}).}$$

Proof. Take $T \in \mathcal{R}_y(B)$, realized from y by P with $c(P) \preceq B$. Let Q take x to y with $c(Q) \preceq c_{xy}$. The concatenated protocol $Q \circ P$ realizes T from x and has cost at most $c_{xy} + B$. Thus $T \in \mathcal{R}_x(B + c_{xy})$. $\square$

Corollary 22 (No endogenous unbudgeted expansion). *With $\mathcal{R}_x^\infty = \bigcup_B \mathcal{R}_x(B)$,*

$$x \rightsquigarrow y \implies \mathcal{R}_y^\infty \subseteq \mathcal{R}_x^\infty.$$

If x and y are mutually reachable, then their unbudgeted reachable task envelopes are equal.

The result is a budget-shifted monotonicity law, not a new reachability theorem. Its GC role is to expose a frequent interpretive error: a capability observed after an internal transition may look newly created only because the resource cost of reaching the later state was omitted.

Algorithm 5 gives the finite audit used in the software.

The earlier unrestricted idea that every closed system must conserve its scalable envelope class is too strong. A finite bijective microscopic map can be operationally expensive to reverse.

Theorem 23 (Scalable Nonconservation under Expensive Reversal). *For every $n \geq 1$, there is a closed finite system with bijective dynamics $U_n(q) = q + 1 \pmod{2^n}$ such that one forward step costs 1, whereas implementing the inverse step using only the admissible forward dynamics costs $2^n - 1$. Consequently, microscopic bijectivity does not imply polynomial-overhead operational reversibility for the family $\{U_n\}_{n \geq 1}$.*

Algorithm 5 Finite budgeted reachable-envelope transport audit

Require: Weighted directed state graph, state-to-task map, maximum budget $B_{\max}$

```
1: for all ordered reachable state pairs  $(x, y)$  do
2:   Compute minimum transition cost  $c_{xy}$ 
3:   for  $B = 0, \dots, B_{\max}$  do
4:     Compute  $\mathcal{R}_y(B)$  and  $\mathcal{R}_x(B + c_{xy})$ 
5:     Record whether  $\mathcal{R}_y(B) \subseteq \mathcal{R}_x(B + c_{xy})$ 
6:   end for
7: end for
8: return all witnesses and inclusion verdicts
```

Proof. U_n is a permutation of $\mathbb{Z}_{2^n}$ and therefore has an inverse. From q to $q + 1$ requires one admissible step. From $q + 1$ back to q using only forward applications requires $2^n - 1$ additional forward steps. The reversal overhead is exponential in n , so it is excluded by any polynomial overhead class. $\square$

Proposition 24 (Operational Reversibility Conservation). *If two stages W_s, W_t admit faithful translations in both directions with overhead in the same composition-closed class $\mathcal{H}$, then $W_s \simeq_G^{\mathcal{H}} W_t$ and hence their scalable envelope classes coincide.*

Proof. This is exactly mutual reachability under the generative preorder. $\square$

10 Programmability connection

The exact quantum no-programming theorem requires orthogonal program states for distinct exactly implemented unitary operations (Nielsen & Chuang, 1997); approximate programmable processors relax this exact setting and already possess independent accuracy–dimension bounds (Hillery et al., 2006). GC therefore treats programmability as a constrained specialization rather than a novelty claim.

Corollary 25 (Exact Generator Realizability). *Suppose a fixed deterministic programmable quantum processor must select exactly among g distinct unitary primitives required by an envelope generator set. Then its program Hilbert space must have dimension at least g . A program register of b qubits therefore requires*

$$b \geq \lceil \log_2 g \rceil.$$

Proof. Nielsen–Chuang exact programmability implies pairwise orthogonal program states for distinct exact unitaries (Nielsen & Chuang, 1997). A Hilbert space containing g mutually orthogonal states has dimension at least g ; b qubits have dimension 2^b . $\square$

11 Reproducible benchmark design

The companion software evaluates eleven benchmark families. Table 2 states their epistemic role before any numerical result appears.

12 Results

12.1 Established-theory validation

Table 3 reports the exact symmetric-binary deficiency generated by the software and recovers the expected Blackwell directionality.

Figure 1 visualizes the same directional deficiency pattern.

Table 2: Benchmark families and epistemic role.

Benchmark	Existing result checked	GC-specific use
Binary symmetric experiments	Blackwell/Le Cam	Exact reduction validation
Two-resource Pareto worlds	Multiobjective feasibility	Marginal incompleteness
Equality communication	Communication complexity	Resource-scaling projection
Polynomial hierarchy slice	Complexity hierarchy	Nested envelope illustration
Finite envelope grid	Enumeration	Whole-envelope logic
Parity envelopes	Marginal/higher-order structure	Proper-projection irreducibility
Convex support path	Convex analysis	Reconstruction theorem validation
Cyclic bijective system	Bijective dynamics	Operational reversal-cost separation
Weighted reachability system	Reachability/viability	Budgeted transport audit
Exact program dimension	Quantum no-programming	Generator realizability corollary
Public MLPerf Power subset	Performance-power benchmarking	Non-toy multi-workload validation

Table 3: Binary symmetric experiment comparison.

Source accuracy	Target accuracy	TV deficiency	Exact simulation
0.75	0.90	0.150	No
0.90	0.75	0.000	Yes
0.60	0.80	0.200	No
0.80	0.60	0.000	Yes
0.70	0.70	0.000	Yes

Table 4 reports the Equality communication benchmark at $\epsilon = 1/3$.

Figure 2 shows the scaling contrast.

12.2 Joint-resource and projection structure

Table 5 verifies the joint-budget discrepancy in the Marginal Incompleteness construction.

Figure 3 displays the two frontiers.

Table 6 summarizes the parity-family result through $m = 8$: every projection below full order agrees, while the whole relations differ.

Figure 4 shows the unbounded finite hierarchy $r_G = m$.

12.3 Reconstruction and dynamic tests

Table 7 shows that numerical support-function integration remains at floating-point error for the exact Minkowski path.

Figure 5 shows numerical error remaining at machine-precision scale.

Table 8 reports the exponential reversal-cost construction.

Figure 6 visualizes the same exponential asymmetry.

The reachable-envelope theorem was exhaustively checked on a five-state weighted system, and Table 9 reports the audit summary.

Figure 7 displays the pass fraction over the tested budget grid.

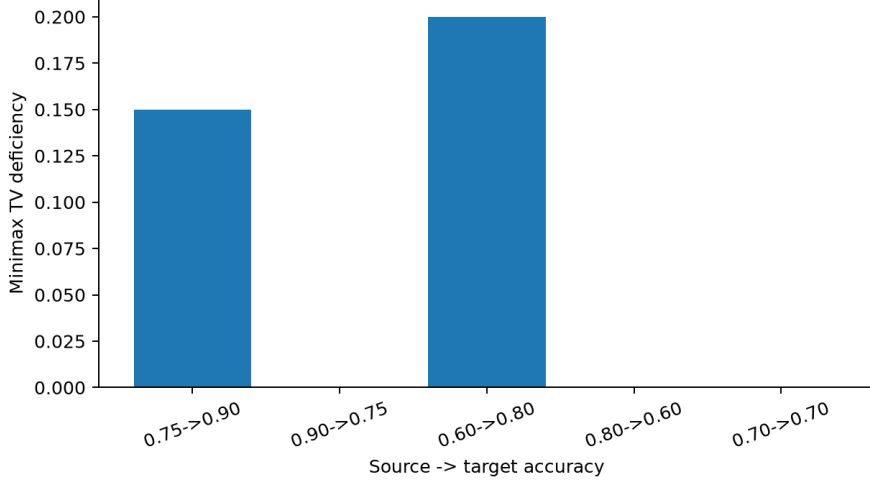


Figure 1: Minimax total-variation deficiency for the symmetric-binary validation benchmark.

Table 4: Equality communication benchmark at $\epsilon = 1/3$.

n	Deterministic bits	Public-randomized proxy	Ratio
8	9	3	3.00
16	17	3	5.67
32	33	3	11.00
64	65	3	21.67
128	129	3	43.00
256	257	3	85.67
512	513	3	171.00
1024	1025	3	341.67

12.4 Public MLPerf Power validation

The final benchmark uses public MLPerf Power measurements rather than constructed costs. MLPerf Power reports 1,841 reproducible measurements from 60 systems and explicitly documents performance–energy trade-offs (Tschand et al., 2025). Table 10 compares two systems reporting both ResNet and BERT-99.0 results.

For a portfolio requiring ResNet throughput at least 150,000 q/s, BERT throughput at least 5,000 q/s, and power at most 1000 W for each workload, the GIGABYTE system passes both throughput thresholds but fails both power constraints; the Krai system passes both power constraints but fails both throughput thresholds. The coordinate-wise synthetic ideal combines GIGABYTE throughput with Krai power and is not attained by either tested system. Figure 8 shows the workload-specific observations.

13 Comparison with neighboring theories

Table 11 states exactly where GC reduces to an existing framework and where it asks a broader scoped question. Multi-object resource theories already combine distinct resource-bearing objects in operational tasks (Ducura et al., 2020); GPT simulations already compare operational theories while preserving equivalences (Catani et al., 2026); and quantum programmability already constrains exact and approximate control (Hillery et al., 2006; Nielsen & Chuang, 1997). These results materially narrow the novelty claim.

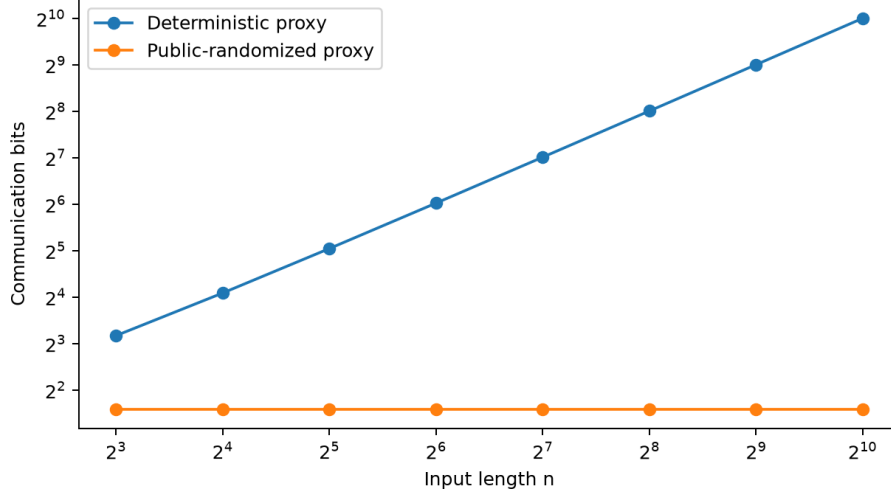


Figure 2: Communication-cost scaling in the Equality validation benchmark.

Table 5: Joint-budget feasibility for the Marginal Incompleteness construction.

Time budget	Energy budget	World A	World B
1	1	No	Yes
1	2	No	Yes
1	3	Yes	Yes
2	1	No	Yes
2	2	No	Yes
2	3	Yes	Yes
3	1	Yes	Yes
3	2	Yes	Yes
3	3	Yes	Yes

14 Discussion and falsifiability

The strongest conclusion is not that nature creates new possibilities. In a fixed operational model, the Budgeted Reachable-Envelope Transport Theorem states the opposite: anything realizable after an internally reachable transition was already realizable before once the transition cost is charged. Thus apparent expansion can be a resource-accounting artifact. Conversely, the cyclic construction shows that microscopic bijectivity is insufficient for scalable operational equivalence; the cost of reversal matters.

The framework therefore separates three statements that are often conflated. First, a formal description may exist while resource-feasible realization fails. Second, a later state may exhibit a task cheaply even though the earlier state could reach it only after paying transition cost. Third, two stages linked by a bijective microscopic map may fail to be equivalent under a restricted overhead class. These distinctions are mathematically testable without asserting new fundamental physics.

Thermodynamic information processing already supplies one-shot work costs, negative-entropy effects, fluctuation relations, multiple-charge constraints, and quantum majorization conditions (Attouch, 1984; Buscemi & Gour, 2017; del Rio et al., 2011; Faist et al., 2015; Gour et al., 2018; Guryanova et al., 2016; Halpern et al., 2016; Sagawa & Ueda, 2012). These mature results further constrain any future attempt to interpret an envelope bound as a new physical law rather than a repackaging of established resource restrictions.

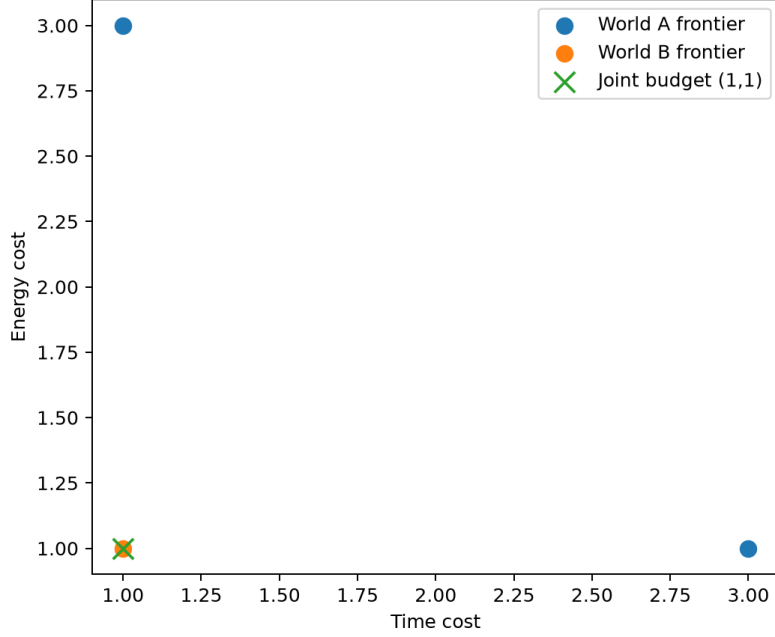


Figure 3: Equal coordinate minima with different joint feasibility.

Table 6: Proper-projection irreducibility and GC projection order.

m	Highest agreeing proper order	r_G	Whole envelopes equal?
2	1	2	No
3	2	3	No
4	3	4	No
5	4	5	No
6	5	6	No
7	6	7	No
8	7	8	No

A breakthrough-level physical claim would require more: a quantitative prediction or bound on whole-envelope behavior that is not already implied by statistical comparison, resource theory, reachability, programmability, complexity, or thermodynamics, followed by an empirical system capable of discriminating the predictions. This manuscript does not claim that threshold has been crossed.

15 Reproducibility and software

The companion Generative Calculus software package is released under the Apache License 2.0. It contains the complete finite-envelope implementation, exact binary-sensor reduction, marginal-incompleteness sweep, Equality and hierarchy validations, proper-projection audit, convex reconstruction test, reversal-cost family, exact program-dimension corollary, reachable-envelope transport audit, public MLPerf Power subset, automated tests, command-line interface, ten-tab Gradio application, generated CSV tables, figures, and machine-readable theorem/novelty reports. GitHub Actions runs tests and full reproduction on pushes, pull requests, and manual dispatch. The audited local run reported 14 passing tests and regenerated all result families deterministically; the clean-environment dependency-download step is delegated to GitHub Actions because the local execution environment has no outbound package-index access.

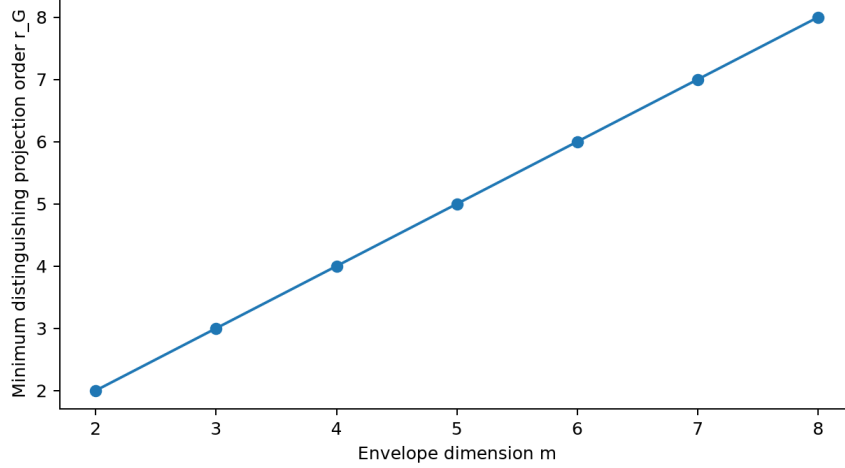


Figure 4: GC projection order for the finite parity-envelope family.

Table 7: Convex support-function reconstruction audit at $t = 0.73$.

Integration steps	Directions	Maximum absolute error
1	720	4.44×10^{-16}
10	720	1.33×10^{-15}
50	720	5.55×10^{-15}
100	720	1.11×10^{-14}
250	720	2.73×10^{-14}

16 Limitations

The theory is deliberately narrower than several early formulations. First, many structural results are exact reductions or applications of established mathematics; their role is integration and consistency, not historical priority. Second, proper-projection irreducibility uses a classical parity construction, so the higher-order combinatorial phenomenon itself is not claimed as new. Third, convex reconstruction relies on standard support-function theory, and reachable-envelope transport relies on protocol concatenation closely related to ordinary reachability. Fourth, the MLPerf experiment validates multi-resource representation but cannot establish a new physical law because performance–power trade-offs are already a mature benchmark concern (Tschand et al., 2025). Finally, no finite literature search can prove universal historical novelty.

The remaining research frontier is therefore sharply stated: identify a natural physical or computational setting in which a whole-envelope theorem yields a quantitative consequence not derivable from the mature neighboring frameworks enumerated above. Until such a consequence is obtained and independently tested, Generative Calculus should be read as a rigorous foundational synthesis rather than an experimentally established revolution.

17 Conclusion

Generative Calculus organizes physically realizable capability through total task–scale–error–budget–composition envelopes. The framework recovers Blackwell–Le Cam comparison, resource convertibility, and complexity specializations; proves representation, preorder, computability, marginal-incompleteness, and proper-projection results; obtains a rigorous support-function reconstruction sector; rejects unrestricted envelope conservation through an explicit reversal-cost construction; and replaces it with a budgeted reachable-envelope transport theorem. Public

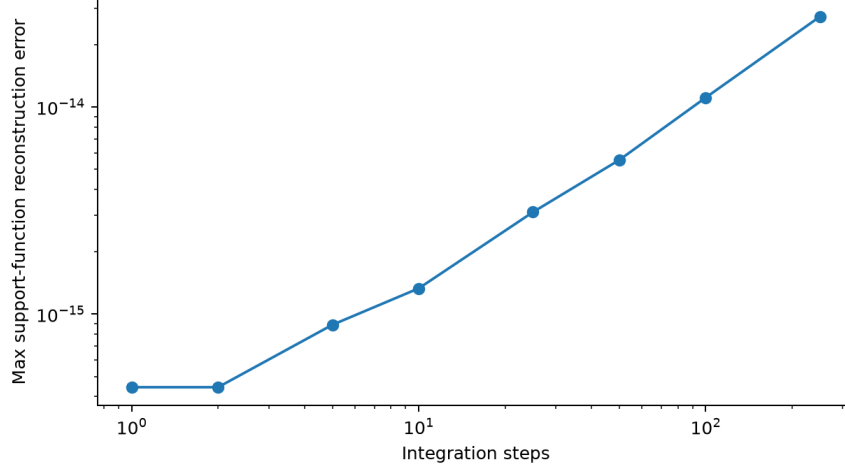


Figure 5: Numerical error in the restricted convex reconstruction experiment.

Table 8: Bijective microscopic cycle with costly operational reversal.

n	Period 2^n	Forward cost	Reverse-via-forward cost
4	16	1	15
8	256	1	255
12	4096	1	4095
16	65536	1	65535
20	1048576	1	1048575

MLPerf Power data provide a non-toy validation of the joint-resource representation. The central conceptual lesson is precise: what appears to become possible later need not have been created later—it may have been reachable earlier at a cost that was not counted. At the same time, microscopic bijectivity does not guarantee scalable capability reversibility. These two statements provide a disciplined basis for further mathematical and empirical work while keeping the novelty claim within what the current evidence supports.

Data and Code Availability

All code, data, experiments, and reproducibility materials supporting this study are openly available at <https://github.com/Arithmetic-Power-Geometry/Generative-Calculus>. The software is released under the Apache License 2.0.

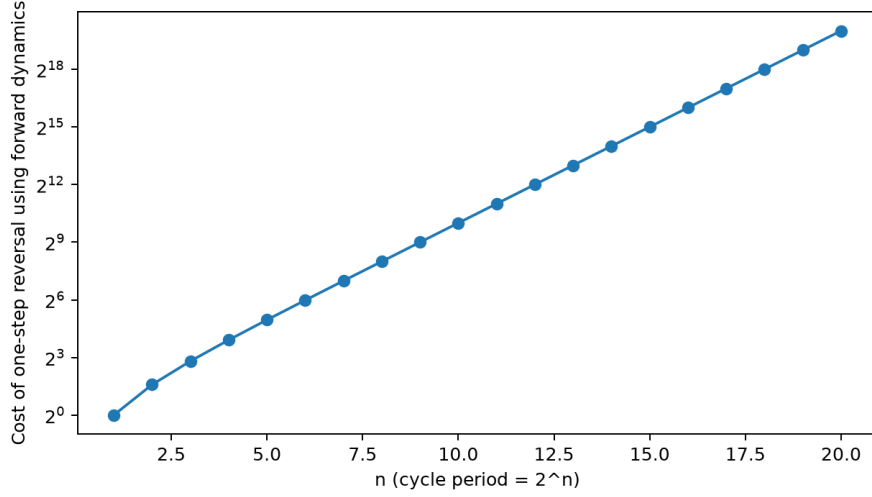


Figure 6: Operational reversal cost under a bijective cyclic microscopic map.

Table 9: Budgeted reachable-envelope transport stress test.

Audit quantity	Value
States	5
Maximum budget	15
Ordered reachable-pair/budget checks	400
Transport inclusions satisfied	400
Violations	0

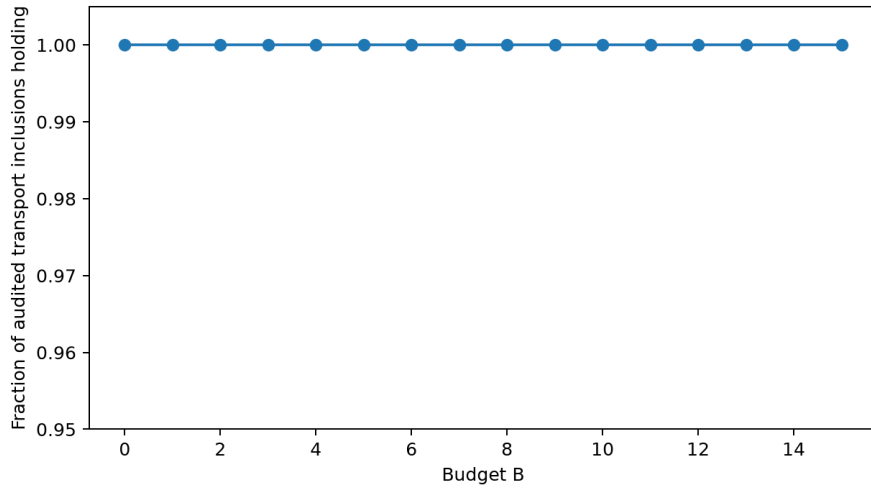


Figure 7: Finite audit of budget-shifted reachable-envelope transport.

Table 10: Public MLPerf Power subset used for the cross-workload portfolio audit.

System/workload	Throughput (q/s)	Power (W)	Accelerators
GIGABYTE G292-Z43 / ResNet	265029.0	1348.775	16
GIGABYTE G292-Z43 / BERT-99.0	9594.98	1361.896	16
Krai R282-Z93 / ResNet	94996.9	484.509	5
Krai R282-Z93 / BERT-99.0	2848.1	424.309	5

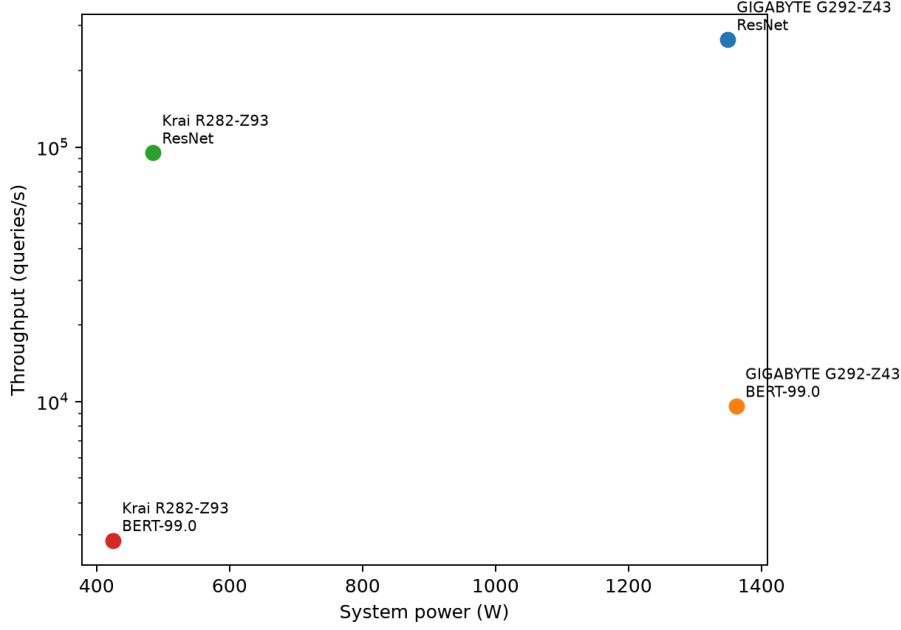


Figure 8: Public MLPerf Power performance–power observations used in the non-toy validation.

Table 11: Existing frameworks versus the scoped Generative Calculus question.

Framework	Established question	GC specialization or extension
Blackwell/Le Cam	Can one experiment simulate another, exactly or approximately?	Exact reduction on the statistical-experiment slice
Resource theory	Can one resource object convert to another under free operations?	Exact reduction on fixed free-operation slices
GPT simulation/contextuality	Can operational theories embed under equivalence-preserving simulations?	Neighboring whole-theory comparison; GC adds explicit scale/error/resource-budget indexing
Multi-object resource theory	Can multiple resources jointly advantage an operational task?	Strong prior art against claiming joint resources as new
Reachability/viability	Which states are reachable or viable under constrained dynamics?	GC transports the logic to task envelopes with explicit accumulated resource cost
Quantum programming	Which exact/approximate operations fit a finite program space?	Specialization constraining envelope generator realizability
Complexity theory	Which tasks are feasible under asymptotic time/space resources?	Computational projection of the envelope formalism
Generative Calculus	Which task–scale–error–budget–composition tuples are jointly feasible, and how do complete envelopes translate along dynamics?	Scoped synthesis studied here

References

- Aaronson, S. (2005). NP-complete problems and physical reality. *SIGACT News*, 36(1), 30–52.
- Abramsky, S., & Coecke, B. (2004). A categorical semantics of quantum protocols. *Proceedings of the 19th Annual IEEE Symposium on Logic in Computer Science*, 415–425. <https://doi.org/10.1109/LICS.2004.1319636>
- Adams, A., Zenil, H., Davies, P. C. W., & Walker, S. I. (2017). Formal definitions of unbounded evolution and innovation reveal universal mechanisms for open-ended evolution in dynamical systems. *Scientific Reports*, 7, 997. <https://doi.org/10.1038/s41598-017-00810-8>
- Ambrosio, L., Gigli, N., & Savaré, G. (2008). *Gradient flows in metric spaces and in the space of probability measures* (2nd ed.). Birkhäuser.
- Arora, S., & Barak, B. (2009). *Computational complexity: A modern approach*. Cambridge University Press.
- Attouch, H. (1984). *Variational convergence for functions and operators*. Pitman.
- Aubin, J.-P. (2005). Viability kernels and capture basins of sets under differential inclusions. *SIAM Journal on Control and Optimization*, 43(3), 853–881. <https://doi.org/10.1137/S036301290036968X>
- Awodey, S. (2010). *Category theory* (2nd ed.). Oxford University Press.
- Babai, L., & Kimmel, P. G. (1997). Randomized simultaneous messages: Solution of a problem of Yao in communication complexity. *Proceedings of the 12th IEEE Conference on Computational Complexity*, 239–246.
- Baez, J. C., & Pollard, B. S. (2017). A compositional framework for reaction networks. *Reviews in Mathematical Physics*, 29, 1750028. <https://doi.org/10.1142/S0129055X17500283>
- Baez, J. C., & Stay, M. (2011). Physics, topology, logic and computation: A rosetta stone. *New Structures for Physics*, 95–172.
- Barnum, H., Barrett, J., Leifer, M., & Wilce, A. (2007). Generalized no-broadcasting theorem. *Physical Review Letters*, 99, 240501. <https://doi.org/10.1103/PhysRevLett.99.240501>
- Barnum, H., Barrett, J., Leifer, M., & Wilce, A. (2010). Teleportation in general probabilistic theories. *Proceedings of Symposia in Applied Mathematics*, 71, 25–48.
- Barnum, H., Müller, M. P., & Ududec, C. (2014). Higher-order interference and single-system postulates characterizing quantum theory. *New Journal of Physics*, 16, 123029. <https://doi.org/10.1088/1367-2630/16/12/123029>
- Barrett, J. (2007). Information processing in generalized probabilistic theories. *Physical Review A*, 75, 032304. <https://doi.org/10.1103/PhysRevA.75.032304>
- Bedau, M. A., McCaskill, J. S., Packard, N. H., Rasmussen, S., Adami, C., Green, D. G., Ikegami, T., Kaneko, K., & Ray, T. S. (2000). Open problems in artificial life. *Artificial Life*, 6(4), 363–376. <https://doi.org/10.1162/106454600300103683>
- Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6), 525–532. <https://doi.org/10.1147/rd.176.0525>
- Bennett, C. H. (1982). The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21, 905–940. <https://doi.org/10.1007/BF02084158>
- Blackwell, D. (1951). Comparison of experiments. *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, 93–102.
- Blackwell, D. (1953). Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2), 265–272. <https://doi.org/10.1214/aoms/1177729032>
- Blum, M. (1967). A machine-independent theory of the complexity of recursive functions. *Journal of the ACM*, 14(2), 322–336. <https://doi.org/10.1145/321386.321395>
- Borg, J. M., Buskell, A., Kapitany, R., Powers, S. T., Reindl, E., & Tennie, C. (2024). Evolved open-endedness in cultural evolution: A new dimension in open-ended evolution research. *Artificial Life*, 30(3), 417–438. https://doi.org/10.1162/artl_a_00406
- Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge University Press.

- Brandão, F. G. S. L., & Gour, G. (2015). Reversible framework for quantum resource theories. *Physical Review Letters*, 115(7), 070503. <https://doi.org/10.1103/PhysRevLett.115.070503>
- Brandão, F. G. S. L., Horodecki, M., Ng, N. H. Y., Oppenheim, J., & Wehner, S. (2015). The second laws of quantum thermodynamics. *Proceedings of the National Academy of Sciences*, 112(11), 3275–3279. <https://doi.org/10.1073/pnas.1411728112>
- Buhrman, H., Cleve, R., Massar, S., & de Wolf, R. (2010). Nonlocality and communication complexity. *Reviews of Modern Physics*, 82, 665–698. <https://doi.org/10.1103/RevModPhys.82.665>
- Burago, D., Burago, Y., & Ivanov, S. (2001). *A course in metric geometry*. American Mathematical Society.
- Buscemi, F. (2012). All entangled quantum states are nonlocal. *Physical Review Letters*, 108, 200401. <https://doi.org/10.1103/PhysRevLett.108.200401>
- Buscemi, F., & Gour, G. (2017). Quantum relative lorenz curves. *Physical Review A*, 95, 012110. <https://doi.org/10.1103/PhysRevA.95.012110>
- Catani, L., Galley, T. D., & Gonda, T. (2026). Resource-theoretic hierarchy of contextuality for general probabilistic theories. *Quantum*, 10, 2077. <https://doi.org/10.22331/q-2026-04-21-2077>
- Chaitin, G. J. (1974). Information-theoretic limitations of formal systems. *Journal of the ACM*, 21(3), 403–424. <https://doi.org/10.1145/321832.321839>
- Channon, A. (2024). A procedure for testing for Tokyo Type 1 open-ended evolution. *Artificial Life*, 30(3), 345–355. https://doi.org/10.1162/artl_a_00430
- Chiribella, G., D’Ariano, G. M., & Perinotti, P. (2010). Probabilistic theories with purification. *Physical Review A*, 81, 062348. <https://doi.org/10.1103/PhysRevA.81.062348>
- Chiribella, G., D’Ariano, G. M., & Perinotti, P. (2011). Informational derivation of quantum theory. *Physical Review A*, 84, 012311. <https://doi.org/10.1103/PhysRevA.84.012311>
- Chitambar, E., & Gour, G. (2019). Quantum resource theories. *Reviews of Modern Physics*, 91(2), 025001. <https://doi.org/10.1103/RevModPhys.91.025001>
- Church, A. (1936). An unsolvable problem of elementary number theory. *American Journal of Mathematics*, 58(2), 345–363. <https://doi.org/10.2307/2371045>
- Clavel, M., Durán, F., Eker, S., Lincoln, P., Martí-Oliet, N., Meseguer, J., & Talcott, C. (2007). *All about maude: A high-performance logical framework*. Springer.
- Coecke, B., & Duncan, R. (2011). Interacting quantum observables: Categorical algebra and diagrammatics. *New Journal of Physics*, 13, 043016. <https://doi.org/10.1088/1367-2630/13/4/043016>
- Coecke, B., Fritz, T., & Spekkens, R. W. (2016). A mathematical theory of resources. *Information and Computation*, 250, 59–86. <https://doi.org/10.1016/j.ic.2016.02.008>
- Coecke, B., & Kissinger, A. (2017). *Picturing quantum processes*. Cambridge University Press.
- Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, 151–158. <https://doi.org/10.1145/800157.805047>
- Cover, T. M., & Thomas, J. A. (2006). *Elements of information theory* (2nd ed.). Wiley.
- Csiszár, I. (1967). Information-type measures of difference of probability distributions and indirect observations. *Studia Scientiarum Mathematicarum Hungarica*, 2, 299–318.
- Deb, K. (2001). *Multi-objective optimization using evolutionary algorithms*. Wiley.
- del Rio, L., Åberg, J., Renner, R., Dahlsten, O., & Vedral, V. (2011). The thermodynamic meaning of negative entropy. *Nature*, 474, 61–63. <https://doi.org/10.1038/nature10123>
- Deutsch, D. (1985). Quantum theory, the church–turing principle and the universal quantum computer. *Proceedings of the Royal Society A*, 400, 97–117. <https://doi.org/10.1098/rspa.1985.0070>

- Deutsch, D. (2013). Constructor theory. *Synthese*, 190, 4331–4359. <https://doi.org/10.1007/s11229-013-0279-z>
- Deutsch, D., & Marletto, C. (2015). Constructor theory of information. *Proceedings of the Royal Society A*, 471, 20140540. <https://doi.org/10.1098/rspa.2014.0540>
- Ducuara, A. F., Lipka-Bartosik, P., & Skrzypczyk, P. (2020). Multiobject operational tasks for convex quantum resource theories of state-measurement pairs. *Physical Review Research*, 2(3), 033374. <https://doi.org/10.1103/PhysRevResearch.2.033374>
- Ehrgott, M. (2005). *Multicriteria optimization* (2nd ed.). Springer.
- Faist, P., Dupuis, F., Oppenheim, J., & Renner, R. (2015). The minimal work cost of information processing. *Nature Communications*, 6, 7669. <https://doi.org/10.1038/ncomms8669>
- Feynman, R. P. (1982). Simulating physics with computers. *International Journal of Theoretical Physics*, 21, 467–488. <https://doi.org/10.1007/BF02650179>
- Fong, B. (2015). Decorated cospans. *Theory and Applications of Categories*, 30, 1096–1120.
- Fong, B., & Spivak, D. I. (2019). *An invitation to applied category theory: Seven sketches in compositionality*. Cambridge University Press.
- Fritz, T. (2017). Resource convertibility and ordered commutative monoids. *Mathematical Structures in Computer Science*, 27(6), 850–938. <https://doi.org/10.1017/S0960129515000444>
- Goldreich, O. (2008). *Computational complexity: A conceptual perspective*. Cambridge University Press.
- Goold, J., Huber, M., Riera, A., del Rio, L., & Skrzypczyk, P. (2016). The role of quantum information in thermodynamics—a topical review. *Journal of Physics A: Mathematical and Theoretical*, 49, 143001. <https://doi.org/10.1088/1751-8113/49/14/143001>
- Gour, G., Jennings, D., Buscemi, F., Duan, R., & Marvian, I. (2018). Quantum majorization and a complete set of entropic conditions for quantum thermodynamics. *Nature Communications*, 9, 5352. <https://doi.org/10.1038/s41467-018-06261-7>
- Gour, G., Müller, M. P., Narasimhachar, V., Spekkens, R. W., & Halpern, N. Y. (2015). The resource theory of informational nonequilibrium in thermodynamics. *Physics Reports*, 583, 1–58. <https://doi.org/10.1016/j.physrep.2015.04.003>
- Guryanova, Y., Popescu, S., Short, A. J., Silva, R., & Skrzypczyk, P. (2016). Thermodynamics of quantum systems with multiple conserved quantities. *Nature Communications*, 7, 12049. <https://doi.org/10.1038/ncomms12049>
- Halpern, N. Y., Faist, P., Oppenheim, J., & Winter, A. (2016). Microcanonical and resource-theoretic derivations of the thermal state of a quantum system with noncommuting charges. *Nature Communications*, 7, 12051. <https://doi.org/10.1038/ncomms12051>
- Hardy, G. H., Littlewood, J. E., & Pólya, G. (1934). *Inequalities*. Cambridge University Press.
- Hardy, L. (2001). Quantum theory from five reasonable axioms.
- Hartmanis, J., & Stearns, R. E. (1965). On the computational complexity of algorithms. *Transactions of the American Mathematical Society*, 117, 285–306. <https://doi.org/10.2307/1994208>
- Hayashi, M., & Yamasaki, H. (2025). The generalized quantum stein’s lemma and the second law of quantum resource theories. *Nature Physics*, 21, 1988–1993. <https://doi.org/10.1038/s41567-025-03047-9>
- Heinosaari, T., Miyadera, T., & Ziman, M. (2016). An invitation to quantum incompatibility. *Journal of Physics A: Mathematical and Theoretical*, 49, 123001. <https://doi.org/10.1088/1751-8113/49/12/123001>
- Hillery, M., Ziman, M., & Bužek, V. (2006). Approximate programmable quantum processors. *Physical Review A*, 73(2), 022345. <https://doi.org/10.1103/PhysRevA.73.022345>
- Hiriart-Urruty, J.-B., & Lemaréchal, C. (2001). *Fundamentals of convex analysis*. Springer.
- Holland, J. H. (1992). *Adaptation in natural and artificial systems* (2nd ed.). MIT Press.
- Hopcroft, J. E., Motwani, R., & Ullman, J. D. (2006). *Introduction to automata theory, languages, and computation* (3rd ed.). Pearson.

- Horodecki, M., & Oppenheim, J. (2013). Fundamental limitations for quantum and nanoscale thermodynamics. *Nature Communications*, 4, 2059. <https://doi.org/10.1038/ncomms3059>
- Horodecki, R., Horodecki, P., Horodecki, M., & Horodecki, K. (2009). Quantum entanglement. *Reviews of Modern Physics*, 81(2), 865–942. <https://doi.org/10.1103/RevModPhys.81.865>
- Ioannou, M., Sekatski, P., Designolle, S., Jones, B. D. M., Uola, R., & Brunner, N. (2022). Simulability of high-dimensional quantum measurements. *Physical Review Letters*, 129, 190401. <https://doi.org/10.1103/PhysRevLett.129.190401>
- Janotta, P., & Hinrichsen, H. (2014). Generalized probability theories: What determines the structure of quantum theory? *Journal of Physics A: Mathematical and Theoretical*, 47, 323001. <https://doi.org/10.1088/1751-8113/47/32/323001>
- Karp, R. M. (1972). Reducibility among combinatorial problems. In *Complexity of computer computations* (pp. 85–103). Plenum Press.
- Kauffman, S., & Roli, A. (2021). The world is not a theorem. *Entropy*, 23(11), 1467. <https://doi.org/10.3390/e23111467>
- Kauffman, S. A. (2000). *Investigations*. Oxford University Press.
- Khan, M. A., & Yu, H. (2025). A simple proof of Blackwell’s theorem on the comparison of experiments for a general state space. *Economics Letters*, 247, 112146. <https://doi.org/10.1016/j.econlet.2024.112146>
- Khanian, Z. B., Bera, M. N., Riera, A., Lewenstein, M., & Winter, A. (2023). Resource theory of heat and work with non-commuting charges. *Annales Henri Poincaré*, 24, 1725–1777. <https://doi.org/10.1007/s00023-022-01254-1>
- Kullback, S., & Leibler, R. A. (1951). On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1), 79–86. <https://doi.org/10.1214/aoms/1177729694>
- Kushilevitz, E., & Nisan, N. (1997). *Communication complexity*. Cambridge University Press.
- Ladner, R. E. (1975). On the structure of polynomial time reducibility. *Journal of the ACM*, 22(1), 155–171. <https://doi.org/10.1145/321864.321877>
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183–191. <https://doi.org/10.1147/rd.53.0183>
- Langton, C. G. (1986). Studying artificial life with cellular automata. *Physica D*, 22, 120–149. [https://doi.org/10.1016/0167-2789\(86\)90237-X](https://doi.org/10.1016/0167-2789(86)90237-X)
- Lawvere, F. W., & Schanuel, S. H. (2009). *Conceptual mathematics: A first introduction to categories* (2nd ed.). Cambridge University Press.
- Le Cam, L. (1964). Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4), 1419–1455.
- Le Cam, L., & Yang, G. L. (2000). *Asymptotics in statistics: Some basic concepts* (2nd ed.). Springer.
- Lee, C. M., & Barrett, J. (2015). Computation in generalised probabilistic theories. *New Journal of Physics*, 17, 083001. <https://doi.org/10.1088/1367-2630/17/8/083001>
- Lenski, R. E., Ofria, C., Pennock, R. T., & Adami, C. (2003). The evolutionary origin of complex features. *Nature*, 423, 139–144. <https://doi.org/10.1038/nature01568>
- Li, M., & Vitányi, P. (2008). *An introduction to kolmogorov complexity and its applications* (3rd ed.). Springer.
- Liu, Y., Wang, Y., & Luo, S. (2024). Quantum resource theory of coding for error correction. *Physical Review A*, 110(3), 032413. <https://doi.org/10.1103/PhysRevA.110.032413>
- Lloyd, S. (2000). Ultimate physical limits to computation. *Nature*, 406, 1047–1054. <https://doi.org/10.1038/35023282>
- Lostaglio, M. (2019). An introductory review of the resource theory approach to thermodynamics. *Reports on Progress in Physics*, 82(11), 114001. <https://doi.org/10.1088/1361-6633/ab46e5>
- Luo, Y. (2025). One-shot manipulation of coherence in dynamic quantum resource theory. *Physical Review A*, 111(2), 022447. <https://doi.org/10.1103/PhysRevA.111.022447>

- Mac Lane, S. (1998). *Categories for the working mathematician* (2nd ed.). Springer.
- Margolus, N., & Levitin, L. B. (1998). The maximum speed of dynamical evolution. *Physica D*, 120, 188–195. [https://doi.org/10.1016/S0167-2789\(98\)00054-2](https://doi.org/10.1016/S0167-2789(98)00054-2)
- Marletto, C. (2015). Constructor theory of life. *Journal of the Royal Society Interface*, 12, 20141226. <https://doi.org/10.1098/rsif.2014.1226>
- Marletto, C. (2021). *The science of can and can't*. Viking.
- Marshall, A. W., Olkin, I., & Arnold, B. C. (2011). *Inequalities: Theory of majorization and its applications* (2nd ed.). Springer.
- Maynard Smith, J. (1982). *Evolution and the theory of games*. Cambridge University Press.
- Meseguer, J. (1992). Conditional rewriting logic as a unified model of concurrency. *Theoretical Computer Science*, 96(1), 73–155. [https://doi.org/10.1016/0304-3975\(92\)90182-F](https://doi.org/10.1016/0304-3975(92)90182-F)
- Mesoudi, A., & Thornton, A. (2018). What is cumulative cultural evolution? *Proceedings of the Royal Society B*, 285, 20180712. <https://doi.org/10.1098/rspb.2018.0712>
- Miettinen, K. (1999). *Nonlinear multiobjective optimization*. Kluwer Academic.
- Newman, I. (1991). Private vs. common random bits in communication complexity. *Information Processing Letters*, 39(2), 67–71. [https://doi.org/10.1016/0020-0190\(91\)90157-D](https://doi.org/10.1016/0020-0190(91)90157-D)
- Nielsen, M. A. (1999). Conditions for a class of entanglement transformations. *Physical Review Letters*, 83(2), 436–439. <https://doi.org/10.1103/PhysRevLett.83.436>
- Nielsen, M. A., & Chuang, I. L. (1997). Programmable quantum gate arrays. *Physical Review Letters*, 79(2), 321–324. <https://doi.org/10.1103/PhysRevLett.79.321>
- Odifreddi, P. (1989). *Classical recursion theory*. North-Holland.
- Packard, N., Bedau, M. A., Channon, A., Ikegami, T., Rasmussen, S., Stanley, K. O., & Taylor, T. (2019a). Open-ended evolution and open-endedness: Editorial introduction to the open-ended evolution i special issue. *Artificial Life*, 25(1), 1–3. https://doi.org/10.1162/artl_e_00282
- Packard, N., Bedau, M. A., Channon, A., Ikegami, T., Rasmussen, S., Stanley, K. O., & Taylor, T. (2019b). An overview of open-ended evolution: Editorial introduction to the open-ended evolution ii special issue. *Artificial Life*, 25(2), 93–103. https://doi.org/10.1162/artl_a_00291
- Papadimitriou, C. H. (1994). *Computational complexity*. Addison-Wesley.
- Pareto, V. (1896). *Cours d'économie politique*. F. Rouge.
- Parrondo, J. M. R., Horowitz, J. M., & Sagawa, T. (2015). Thermodynamics of information. *Nature Physics*, 11, 131–139. <https://doi.org/10.1038/nphys3230>
- Pinsker, M. S. (1964). *Information and information stability of random variables and processes*. Holden-Day.
- Plávala, M. (2021). General probabilistic theories: An introduction. *Physics Reports*, 931, 1–68. <https://doi.org/10.1016/j.physrep.2021.06.001>
- Polyanskiy, Y., & Wu, Y. (2025). *Information theory: From coding to learning*. Cambridge University Press. <https://doi.org/10.1017/9781108966351>
- Reeb, D., & Wolf, M. M. (2014). An improved Landauer principle with finite-size corrections. *New Journal of Physics*, 16, 103011. <https://doi.org/10.1088/1367-2630/16/10/103011>
- Regula, B. (2018). Convex geometry of quantum resource quantification. *Journal of Physics A: Mathematical and Theoretical*, 51(4), 045303. <https://doi.org/10.1088/1751-8121/aa9100>
- Regula, B., & Lami, L. (2024). Reversibility of quantum resources through probabilistic protocols. *Nature Communications*, 15, 3096. <https://doi.org/10.1038/s41467-024-47243-2>
- Rice, H. G. (1953). Classes of recursively enumerable sets and their decision problems. *Transactions of the American Mathematical Society*, 74, 358–366. <https://doi.org/10.2307/1990888>
- Rockafellar, R. T. (1970). *Convex analysis*. Princeton University Press.
- Rockafellar, R. T., & Wets, R. J.-B. (1998). *Variational analysis*. Springer.
- Rogers, H. (1987). *Theory of recursive functions and effective computability*. MIT Press.

- Sagawa, T., & Ueda, M. (2012). Nonequilibrium thermodynamics of feedback control. *Physical Review E*, 85, 021104. <https://doi.org/10.1103/PhysRevE.85.021104>
- Sayama, H., & Nehaniv, C. L. (2025). Self-reproduction and evolution in cellular automata: 25 years after evoloops. *Artificial Life*, 31(1), 81–95. https://doi.org/10.1162/artl_a_00451
- Schmid, D., Spekkens, R. W., & Wolfe, E. (2018). All the noncontextuality inequalities for arbitrary prepare-and-measure experiments with respect to any fixed set of operational equivalences. *Physical Review A*, 97, 062103. <https://doi.org/10.1103/PhysRevA.97.062103>
- Schneider, R. (2014). *Convex bodies: The Brunn–Minkowski theory* (2nd ed.). Cambridge University Press. <https://doi.org/10.1017/CBO9781139003858>
- Selinger, P. (2011). A survey of graphical languages for monoidal categories. In *New structures for physics* (pp. 289–355). Springer.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379–423, 623–656.
- Short, A. J., & Wehner, S. (2010). Entropy in general physical theories. *New Journal of Physics*, 12, 033023. <https://doi.org/10.1088/1367-2630/12/3/033023>
- Siefert, J. A., Bird, T. J., Thompson, A. F., Glunt, J. J., Koeln, J. P., Jain, N., & Pangborn, H. C. (2025). Reachability analysis using hybrid zonotopes and functional decomposition. *IEEE Transactions on Automatic Control*, 70(7), 4671–4686. <https://doi.org/10.1109/TAC.2025.3528352>
- Sipser, M. (2012). *Introduction to the theory of computation* (3rd ed.). Cengage Learning.
- Spekkens, R. W. (2005). Contextuality for preparations, transformations, and unsharp measurements. *Physical Review A*, 71, 052108. <https://doi.org/10.1103/PhysRevA.71.052108>
- Strasser, H. (1985). *Mathematical theory of statistics: Statistical experiments and asymptotic decision theory*. de Gruyter.
- Takagi, R., & Regula, B. (2019). General resource theories in quantum mechanics and beyond: Operational characterization via discrimination tasks. *Physical Review X*, 9(3), 031053. <https://doi.org/10.1103/PhysRevX.9.031053>
- Taylor, T. (2019). Evolutionary innovations and where to find them: Routes to open-ended evolution in natural and artificial systems. *Artificial Life*, 25(2), 207–224. https://doi.org/10.1162/artl_a_00290
- Taylor, T., Bedau, M., Channon, A., Ackley, D., Banzhaf, W., Beslon, G., Dolson, E., Froese, T., Hickinbotham, S., Ikegami, T., et al. (2016). Open-ended evolution: Perspectives from the OEE workshop in York. *Artificial Life*, 22(3), 408–423. https://doi.org/10.1162/ARTL_A_00210
- Torgersen, E. (1991). *Comparison of statistical experiments*. Cambridge University Press.
- Tschand, A., Rajan, A. T. R., Idgunji, S., Ghosh, A., Holleman, J., Kiraly, C., Ambalkar, P., Borkar, R., Chukka, R., Cockrell, T., Curtis, O., Fursin, G., Hodak, M., Kassa, H., Lokhmotov, A., Miskovic, D., Pan, Y., Manmathan, M. P., Raymond, L., ... Reddi, V. J. (2025). MLPerf Power: Benchmarking the energy efficiency of machine learning systems from microwatts to megawatts for sustainable AI. *2025 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 1201–1216. <https://doi.org/10.1109/HPCA61900.2025.00092>
- Tsybakov, A. B. (2009). *Introduction to nonparametric estimation*. Springer.
- Turing, A. M. (1936). On computable numbers, with an application to the entscheidungsproblem. *Proceedings of the London Mathematical Society*, s2-42(1), 230–265. <https://doi.org/10.1112/plms/s2-42.1.230>
- van der Vaart, A. W. (1998). *Asymptotic statistics*. Cambridge University Press.
- Villani, C. (2009). *Optimal transport: Old and new*. Springer.
- Wald, A. (1950). *Statistical decision functions*. Wiley.

- Wolpert, D. H. (2019). The stochastic thermodynamics of computation. *Journal of Physics A: Mathematical and Theoretical*, 52, 193001. <https://doi.org/10.1088/1751-8121/ab0850>
- Yang, B. (2025). Emergence of self-replicating hierarchical structures in a binary cellular automaton. *Artificial Life*, 31(1), 96–105. https://doi.org/10.1162/artl_a_00449
- Yao, A. C.-C. (1979). Some complexity questions related to distributive computing. *Proceedings of the 11th Annual ACM Symposium on Theory of Computing*, 209–213. <https://doi.org/10.1145/800135.804414>
- Zitzler, E., & Thiele, L. (1999). Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach. *IEEE Transactions on Evolutionary Computation*, 3(4), 257–271. <https://doi.org/10.1109/4235.797969>